

Fig. 3: Traffic baton for LED flasher

flashing bulb.

Assemble the LED flasher and bulb flasher circuits on separate general-purpose PCBs. Enclose the LED flasher in a transparent acrylic pipe as shown in Fig. 3. The bulb flasher can be enclosed in another transparent acrylic pipe as shown in Fig. 4. Slide switches and red and green acrylic sheets are used for appropriate colour emissions. Now your traffic baton is ready to use. ●

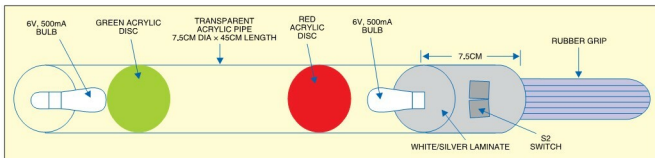


Fig. 4: Traffic baton for bulb flasher

## SHOCK-HAZARD WARNING

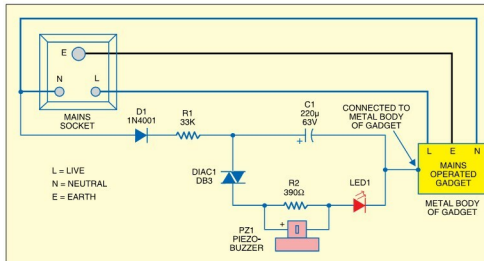
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**E**lectrical leakage can cause lethal shocks. But such an unfortunate situation can be avoided with this shock-hazard warning system. It uses minimal number of components

and does not need any separate power supply.

The circuit continuously monitors the leakage from an electronics gadget by sensing the potential in its body. When there is leakage from the body, the potential exists across the neutral

end and the body of the gadget, which charges capacitor C1 through diode D1 and resistor R1. When the voltage across capacitor C1 exceeds the breakdown potential of DIAC1 (DB3), it starts conducting and the capacitor discharges through the buzzer as well as LED1 quickly. This discharge energy makes LED1 flash and piezobuzzer sound momentarily. As long as there is leakage from the body of the gadget, there is intermittent warning from the buzzer as well as from the LED.



Assemble the circuit on a general-purpose PCB and connect it to the mains-operated gadget. This circuit can be used for washing machines, refrigerators and other similar mains-operated devices. ●